

# Yash Ahuja

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## Education

### Georgia Institute of Technology

M.S. in Mechanical Engineering, Woodruff School — GPA: 3.94/4.0

B.S. in Mechanical Engineering, Woodruff School

Atlanta, GA

Jan 2026 – Dec 2026 (Expected)

Aug 2021 – Dec 2025

## Work Experience

### Andersen Corporation

Strategic Quality Intern

Locust Grove, GA

May – Aug 2025

- Implemented KEYENCE 3D scanning workflows for prototype inspection, dimensional validation, and manufacturing quality control.
- Spearheaded Site Acceptance Testing for \$100K+ equipment, verifying CpK indices to validate process capability and reliability.
- Developed and validated a surface energy testing protocol using Dyne Solution, reducing inspection cycle time by 63% while improving manufacturing throughput.
- Integrated lab devices with PLEX QMS, reducing operator error and enhancing data traceability across the production line.

### Husco International

Mechanical Design Engineering Co-Op, Advanced Vehicle Systems

Waukesha, WI

May – Dec 2023

- Developed and evaluated four valve actuation retention concepts for a CVTMS module in a BEV platform, comparing packaging, manufacturability, and cost tradeoffs to guide design selection.
- Designed, fabricated, and validated endurance test fixtures using GD&T and parametric CAD, with hands-on machining (mill, lathe) supporting prototype development and design validation.
- Conducted CFD simulations reducing pressure drop by 11% through the flow path, improving overall system energy efficiency.
- Performed FEA to validate leaf spring response under normal loads, correlating simulation results with hand calculations and spring rate tester data; modeled bending fixtures accounting for material springback.
- Investigated torque discrepancies between AMESim predictions and system level testing, identifying speed-dependent friction effects and improving model fidelity.

## Research Experience

### Multi-Sensor Fusion Cable Tracking System | Georgia Institute of Technology

Jan - Present 2026

- Designed, assembled, and validated a six-sensor LiDAR test platform integrating custom mechanical mounts, embedded electronics, actuation hardware, and calibration procedures.
- Replaced legacy Siemens motor with ClearPath SD industrial servo on a Rexroth linear rail, enabling precise step-based control (800 steps/rev) and improving motion reliability.
- Engineered fail-safe actuation logic using reed switches and Arduino, preventing hard-stop collisions while preserving full carriage travel.
- Developed embedded C++ firmware on Teensy 4.x microcontrollers interfacing six LD19 2D LiDARs, implementing FOV filtering (220°–320°) to reduce structural noise and false detections.
- Built a real-time multi-cable tracking pipeline (K-means clustering + Hungarian data association + Kalman filtering) for segmentation, identity tracking, and trajectory smoothing under occlusion.

## Projects

### Invision — 3D Coordinate Wall Scanner | Atlanta, GA

Aug – Nov 2025

- Designed a handheld non-destructive wall scanner integrating an 18-antenna UWB radar array (3.3–10 GHz), taking the product from concept through prototype validation while achieving 1 cm volumetric resolution.
- Engineered ToF-based localization system (4× VL53L1X) to anchor radar data to a global frame, eliminating IMU drift.
- Designed and fabricated a handheld consumer hardware enclosure integrating UWB radar, ToF sensors, power electronics, and embedded compute while balancing mechanical packaging, portability, and industrial design requirements.
- Implemented signal-processing-based material classification (wood/metal/PVC) using amplitude filtering and depth thresholding.
- Validated system across 30+ scans, achieving 0.59 in average stud-center error in real-world wall assemblies.

### Smart Robot Manufacturing System | Atlanta, GA

Aug – Nov 2025

- Engineered autonomous pick-and-place system using a UR5 robotic arm controlled via Python RTDE on a Raspberry Pi, integrating custom gripper and vision pipeline.
- Trained custom YOLO model (Ultralytics/Roboflow) with OpenCV for real-time color detection in a manufacturing workflow.
- Developed learning-based mapping (PyTorch MLP) from camera coordinates to robot joint space, reducing manual programming time by 80%.

### HyTech Racing — FSAE Chassis Sub-Team | Atlanta, GA

Jan - Aug 2022

- Fabricated Impact Attenuator test fixture; conducted normal force tests and structured displacement data for the design round of FSAE competition.

## Skills

**CAD / Product Design:** SolidWorks, CATIA, Creo, OnShape, AutoCAD, Product Design, GD&T (ASME Y14.5), Tolerance Stackups, DFx (DFM/DFA), Engineering Drawings

**Simulation / Validation:** ANSYS (FEA & CFD), MATLAB, AMESim, Design of Experiments (DoE), Statistical Process Control (CpK), Python (PyTorch, OpenCV, FastAPI, RTDE), Java

**Rapid Prototyping & Fabrication:** Machining, Welding, Waterjet, Lathe, Bandsaw, Laser Cutter, 3D Printing (FDM & SLA) | PLEX QMS, Windchill PLM, Dewesoft, KEYENCE